

Propagation of olfactory information in *Drosophila*

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Investigating how information propagates between layers in the olfactory system is an important step toward understanding the olfactory code. Each glomerular output projection neuron (PN) receives two sources of input: the olfactory receptor neurons (ORNs) of the same glomerulus and interneurons that innervate many glomeruli. We therefore asked how these inputs interact to produce PN output. We used receptor gene mutations to silence all of the ORNs innervating a specific glomerulus and recorded PN activity with two-photon calcium imaging and electrophysiology. We found evidence for balanced excitatory and inhibitory synaptic inputs but saw little or no response in the absence of direct ORN input. We next asked whether any transformation of activity occurs at successive layers of the antennal lobe. We found a strong link between PN firing and dendritic calcium elevation, the latter of which is tightly correlated with calcium activity in ORN axons, supporting the idea of glomerular propagation of olfactory information. Finally, we showed that odors are represented by a sparse population of PNs. Together, these results are consistent with the idea that direct receptor input provides the main excitatory drive to PNs, whereas interneurons modulate PN output. Balanced excitatory and inhibitory interneuron input may provide a mechanism to adjust PN sensitivity.

antennal lobe | gain modulation | olfaction | smell | sparse code

Sensory systems have the difficult task of encoding the identity and intensity of behaviorally relevant stimuli in the environment. The question of how these stimuli are encoded and propagated from the periphery to higher brain centers is central to systems neuroscience. The stereotypic organization of the *Drosophila* olfactory system and the identification of the odorant receptor genes make the fly an attractive model system in which to study the successive processing of sensory information.

Drosophila olfactory receptor neurons (ORNs) in the antennae detect odors and relay neural activity to the antennal lobe in the brain. An adult fly expresses ≈ 50 odorant receptor genes (1–6). In the antennal lobe, axons of ORNs expressing the same receptor gene project with precision to spatially invariant glomeruli (4–7). Most second-order projection neurons (PNs) send dendrites to individual glomeruli in the antennal lobe and send axons to the mushroom body and the lateral horn of the protocerebrum (8–11).

Imaging experiments in the insect antennal lobe (12–14) reveal a spatial map of glomerular activity. It has been hypothesized that each glomerulus is a functional unit and that the pattern of glomerular activity encodes the quality of odors (15). However, the *Drosophila* antennal lobe contains many GABAergic inhibitory interneurons (10, 16, 17), as well as a group of recently identified cholinergic excitatory interneurons (18). These two classes of interneurons apparently have opposing effects on PNs, and the extent to which they contribute to PN output is unknown. There are two general models for the function of local interneurons in olfactory coding. In one scenario, the interglomerular connections merely modulate the PN response to odors. In this glomerular propagation model, the main source of PN excitation comes from the ORNs of the same glomerulus (its cognate ORNs). In another scenario, lateral and receptor inputs are two independent sources of strong excitatory input for each PN. A powerful lateral input would give rise to a distributed representation of olfactory information in PNs.

To discriminate between these two models, it is necessary to dissect the relative contributions of ORNs and interneurons to PN output. We used receptor gene mutations to silence all of the ORNs innervating a specific glomerulus. Recordings from the cognate PNs in the mutant flies revealed evidence of balanced excitatory and inhibitory synaptic inputs. Nevertheless, we found that PN excitation comes mostly from the cognate receptor input. Furthermore, we asked whether any transformation of activity occurs at various levels of the antennal lobe. Using concurrent calcium imaging and electrophysiology, we found that robust firing in PNs is correlated with dendritic calcium elevation, which is correlated with calcium activity in ORN axons. Finally, we randomly recorded from a population of PNs and found that the representation of odors was sparse. These results are consistent with the idea that direct ORN input provides the main excitatory drive to PNs, whereas excitatory and inhibitory interneurons modulate PN output.

Results

Projection Neuron Output in the Absence of Direct Receptor Input.

Every glomerular output PN receives two sources of synaptic input: the cognate ORNs of the same glomerulus and interneurons that innervate multiple glomeruli (8). What is the main source of neural input to PNs? We reasoned that measuring action potentials of cognate PNs in mutant flies with silenced ORNs should allow us to address this question. If PN activity comes only from the cognate ORNs, it will be eliminated when the ORNs are silenced by receptor gene mutations. Conversely, if some PN activity derives from interglomerular connections, we should still see odor-evoked activity in the PNs as a result of the activation of other glomeruli. Genetic tools are available to perform this experiment in the VM2 glomerulus, which is innervated by ORNs that express the *Or43b* gene (5, 6). With a targeted mutation of the *Or43b* gene (19) and the NP5103 enhancer trap line that labels just the two VM2 PNs (20), we can investigate PN firing properties in the absence of direct ORN input.

We used noninvasive loose-patch recording to measure action potentials in PNs. The VM2 PNs were identified with a fluorescent microscope in flies bearing the *NP5103-Gal4* and *UAS-GFP* transgenes. Two odorants, isoamyl acetate and hexanol, each of which excites different sets of glomeruli in the antennal lobe, were administered at low, medium, and high concentrations to cover the dynamic range of VM2 PNs. At low concentrations, these odorants activate a relatively sparse pattern of glomeruli, and as concentration is increased, a larger number of glomeruli respond. At medium and low concentrations, both isoamyl acetate and hexanol elicit a robust response in VM2 PNs of wild-type flies. In contrast, at these

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Abbreviations: PN, projection neuron; ORN, olfactory receptor neuron; SV, saturated vapor concentration.

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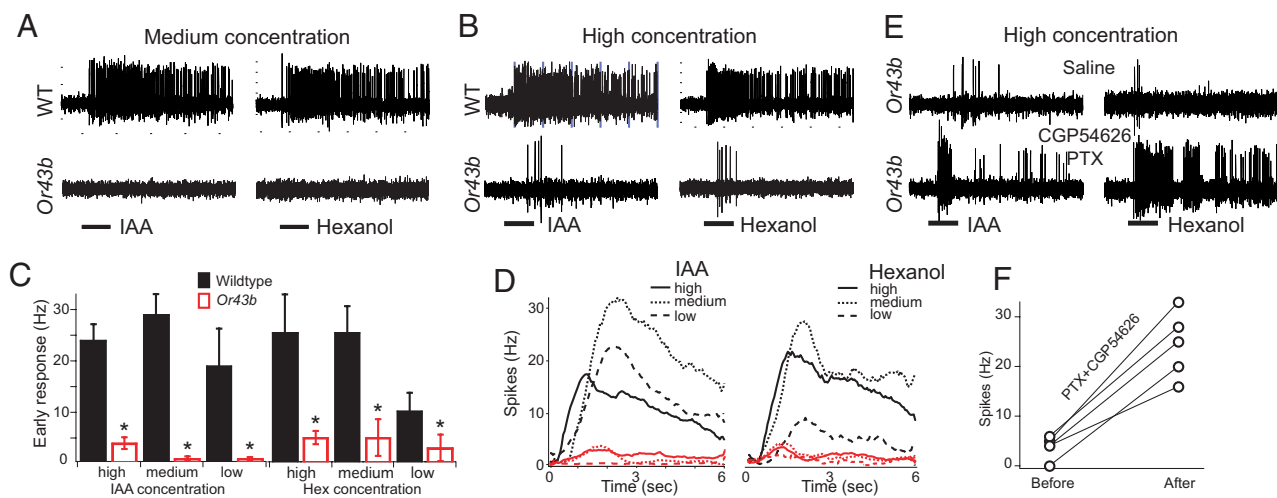


Fig. 1. Cognitive receptor neurons provide most of the input to the action-potential firing of the glomerular output PNs. (A and B) Representative traces of VM2 PNs in response to isoamyl acetate (IAA) and 1-hexanol, respectively, in wild-type and *Or43b* mutant flies at medium (A) and high (B) concentrations. Horizontal bars, 1 s of odor application. (C) Number of action potentials in the first second of response. (D) Averaged instantaneous firing frequency of VM2 PNs to isoamyl acetate and 1-hexanol in wild-type (black lines) and *Or43b* mutant flies (red lines). Frequency was calculated in 50-ms bins and smoothed with a Gaussian filter. $n = 4-9$. (E) Representative traces for before (Upper) and after (Lower) addition of both picrotoxin (PTX; 125 μ M) and CGP54626 (25 μ M). (F) Effect of GABA receptor blockers on the odor-evoked response in VM2 PNs from *Or43b* mutants. Isoamyl acetate, $n = 2$; hexanol, $n = 3$. Odorants were diluted in saline and delivered through a pressure injection system. Isoamyl acetate: high = 1 μ l/ml, medium = 50 nl/ml, and low = 12 nl/ml. Hexanol: high = 5 μ l/ml, medium = 1 μ l/ml, and low = 50 nl/ml. The medium concentration of isoamyl acetate in this experiment was $\approx 8.8\%$ saturated vapor concentration (SV), determined by a bioassay [see supporting information (SI) Methods]. All data are presented as mean $\pm$ SD unless otherwise noted. *, $P < 0.05$; **, $P < 0.01$.

concentrations, VM2 PNs in the *Or43b* mutant flies exhibited little or no detectable response to odor stimulation (Fig. 1A). High concentrations of both odorants did elicit a small response (Fig. 1B); however, the wild-type response was five to six times greater than that of the mutant (Fig. 1C). The dramatic difference in PN firing between wild-type and mutant flies suggests that cognate receptor input makes a greater contribution to the PN firing than any interglomerular connections do. However, the residual excitatory response in the absence of direct receptor input indicates that PNs can be excited by lateral interactions.

It is interesting to note that the PN response in wild-type flies decreased at the high odor concentrations, particularly for isoamyl acetate (Fig. 1D). This finding is consistent with the idea that a larger number of activated glomeruli may induce more lateral inhibition. Indeed, it has been shown that blocking GABA receptors increases PN firing in response to odors (10), demonstrating a role for lateral inhibition in the antennal lobe. The prevalence of lateral inhibition in these conditions raises the question of whether inhibition may be masking the full extent of lateral excitation. Thus, we asked whether blocking inhibition would increase the residual excitatory response observed in the *Or43b* mutant flies. In the presence of picrotoxin and CGP54626, which block GABA_A and GABA_B receptors, respectively, the PN response increased dramatically (Fig. 1E). For isoamyl acetate, we observed an increase from an average of 4 to 23 spikes in the first second ($n = 2$), and from 5 ± 1 to 23 ± 9 spikes ($n = 3$) for hexanol, revealing a strong lateral excitatory connection. Thus, the small residual response observed in Fig. 1B reflects a balance between opposing excitatory and inhibitory lateral interactions.

To rule out the possibility that the residual PN response in the mutant flies was because of ORN activation, we measured odor response of *Or43b* ORNs under the same conditions. With extracellular electrical recordings of single bristles, we verified that the *Or43b* ORNs of mutant flies did not respond to isoamyl acetate at the high concentration, whereas wild-type *Or43b* ORNs responded robustly (SI Fig. 7). Thus, the residual response we observed in VM2 PNs of mutant flies does not derive from the cognate ORNs.

There is a possibility that synaptic efficacy in the VM2 glomer-

ulus is altered by the mutation; for example, in the fly neuromuscular junction, activity influences synapse function (21). To measure synaptic function in mutants, we measured calcium activity in PNs while electrically stimulating the olfactory nerve (13). Two-photon microscopy was used to measure calcium activity in flies bearing *GH146-Gal4* (8) and *UAS-GCaMP* (13) transgenes, in which the calcium sensor G-CaMP (22) is expressed in many PNs. Calcium activity in the dendrites of the VM2 PNs of the *Or43b* mutant flies was indistinguishable from that of wild-type flies (SI Fig. 7E-G), suggesting that a lack of odor-evoked activity does not change the efficacy of the synapse between the *Or43b* ORNs and the VM2 PNs.

Calcium Activity in Cognate and Noncognate PNs in the Absence of Direct Receptor-Neuron Input.

We next investigated whether silencing the *Or43b* ORNs affects the odor response of PNs that innervate other glomeruli. If the impact of lateral excitation is substantial, we should be able to detect a reduction in calcium activity in many glomeruli when input to one glomerulus is removed. We monitored PN dendritic calcium by imaging flies bearing *GH146-Gal4* and *UAS-GCaMP* with two-photon microscopy. This technique allows us to record activity in the entire antennal lobe. Insect nicotinic acetylcholine receptors are highly permeable to calcium (23, 24); therefore monitoring intracellular calcium provides a measure of synaptic excitation. Fig. 2 shows the typical response to odor stimulation in one optical plane of the antennal lobe. In wild-type flies, 4-heptanol, 1-hexen-3-ol, and isoamyl acetate each excite one, two, and four glomeruli in this optical plane (Fig. 2C, E, and G). The VM2 glomerulus of the *Or43b* mutant flies did not show any detectable change of intracellular calcium in response to these three different odorants (Fig. 2D, F, and H). The noncognate glomeruli (DL1, DM3, DM2, and DC2) of this optical plane showed little or no reduction in odor response in the mutant flies compared with the wild-type flies. The DM2 glomerulus showed 19% reduction in response to isoamyl acetate, suggesting that the *Or43b* ORNs may make a small contribution to the activity in noncognate PNs. Results from these imaging experiments corroborate those from electrical recordings

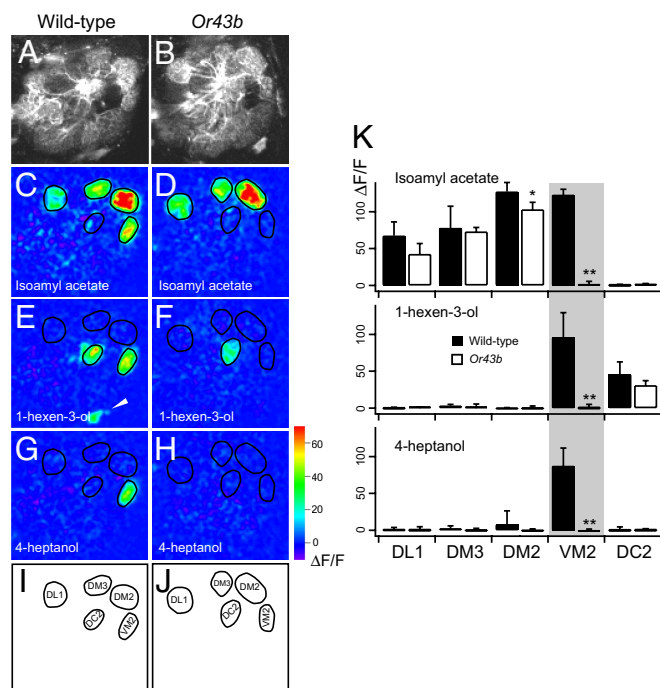


Fig. 2. *Or43b* mutation eliminates odor-evoked calcium activity only in VM2 PNs and has little or no effect on other glomeruli. (A and B) Prestimulation images show glomerular structure. The average of 10 frames before odor exposure is shown. (C–H) Glomerular responses to the three different odors are compared between the wild type and the *Or43b* mutant (C, E, and G versus D, F, and H). Arrowhead in E points to the cell body of a PN with dendrites in a different optical plane. Odor concentration = 8% SV. (I and J) Specific glomeruli were identified anatomically by using the established antennal lobe map (40). (K) Statistical analysis of glomerular response. The olfactory responses in five different glomeruli (DL1, DM3, DM2, VM2, and DC2) were compared between the wild type and the *Or43b* mutant. $\Delta F/F$, fluorescence change. The mutant causes significant change of odor response in the VM2 and DM2 PNs. $n = 4$. *, $p < 0.05$; **, $p < 0.01$.

from the VM2 PNs, suggesting that the cognate ORNs make a much greater contribution to the PN firing than any interglomerular connections do.

Can these conclusions be expanded to other glomeruli? The DM1 glomerulus is innervated by ORNs that express the *Or42b* gene (5, 6). We used a fly line with a *P*-element insertion in the *Or42b* gene (25). Single-sensillum recordings showed that the *Or42b* neurons did not respond to high concentrations of isoamyl acetate (SI Fig. 8 L–N). With calcium imaging, we found that

silencing the *Or42b* ORNs dramatically affected the odor response of the cognate PNs and had little or no influence on the noncognate PNs. In wild-type flies, 3-octanone, benzyl acetate, and isoamyl acetate each evoked different levels of calcium activity in the DM1, DM4, and DP1m glomeruli (SI Fig. 8 C, E, and G). The DM1 glomerulus in the *Or42b* mutant flies did not show any detectable response to these three different odorants (SI Fig. 8 D, F, and H). A quantitative analysis showed that the noncognate glomeruli DM4 and DP1m did not show any significant difference in odor response between the mutant and wild-type flies. These results suggest that the *Or42b* ORNs make the major contribution to the synaptic activity of the DM1 PNs, but make little or no contribution to noncognate PNs.

Calcium Activity in ORNs and PNs of the Same Glomerulus. Thus far, we have shown that lateral inhibitory inputs counterbalance lateral excitation in the VM2 PNs, and that direct ORN input is the main driving force behind PN excitation for VM2 and DM1. If this is a general principle for antennal lobe processing, the responses of PNs innervating other glomeruli should be well correlated with those of their cognate ORNs. To test this hypothesis, we performed quantitative imaging experiments to measure odor-evoked calcium activity in PN dendrites and ORN axons that innervate the same glomerulus. We imaged the *Or43a* ORN axons, which project to the DA4 glomerulus (4–6), and compared the odor-evoked response to that of the DA4 PN dendrites. We measured odor-evoked Ca^{2+} activity in ORN axons by measuring $\Delta F/F$ in the DA4 glomerulus in flies bearing the *Or43a-Gal4* and *UAS-GCaMP* transgenes. These values were compared with those of the DA4 PN dendrites by imaging flies with both *GHI46-Gal4* and *UAS-GCaMP* transgenes (Fig. 3). We observed a strong correlation in $\Delta F/F$ between ORNs and PNs in response to 15 different odorants at each of three concentrations. Similar results were obtained for the DM2 glomerulus (SI Fig. 9), which is innervated by the *Or22a* ORNs (4–6) and responds to a larger repertoire of odorants (26). These results suggest that excitation of PN dendrites results mainly from direct receptor input rather than from lateral connections between glomeruli.

Calcium Activity in Dendrites and Cell Bodies of Individual PNs. In the previous results, we showed that ORN axon activity is well correlated with PN dendritic activity. However, calcium activity in PN dendrites may not accurately reflect PN output to higher brain centers. Calcium activity in PN cell bodies should be a better measure of antennal lobe output. To investigate the relationship between activity in the dendrites and cell body of a given PN, we performed concurrent imaging of PN dendrites and cell bodies. We used the FLP-out technique (27) to randomly label single PNs with G-CaMP, allowing us to image the dendrites and cell body simul-

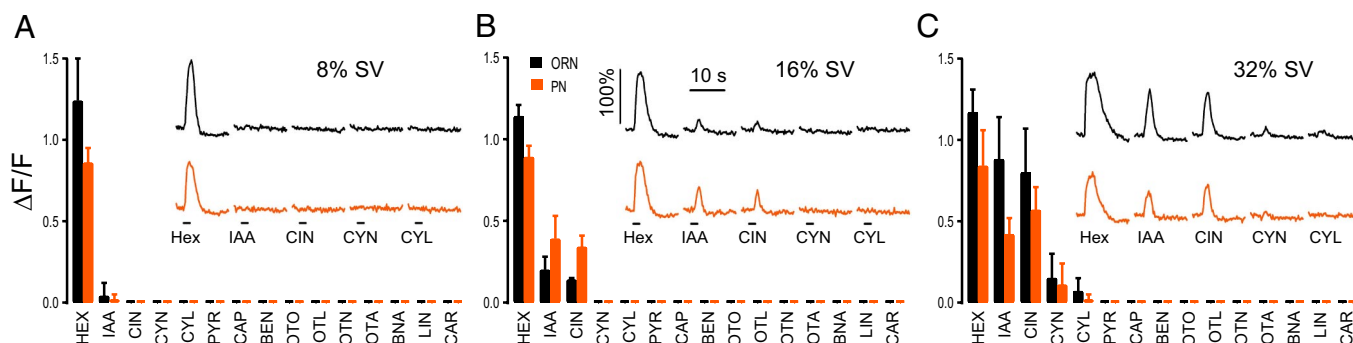


Fig. 3. Concordance in calcium activity between ORNs and PNs of the DA4 glomerulus. Black bars and traces indicate ORNs, and orange bars and traces indicate PNs. Fifteen different odorants (see SI Methods for odor abbreviations) at three different concentrations (8% SV in A, 16% SV in B, and 32% SV in C) were tested. Odor stimulation lasted for 2 s, indicated by the black horizontal lines.

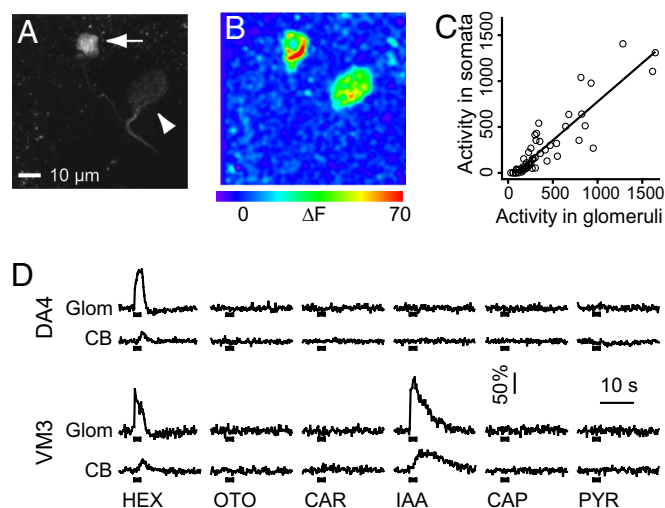


Fig. 4. Ca^{2+} imaging of PN soma and their associated glomeruli. (A) A single PN expressing G-CaMP allows visualization of the soma (arrow) and the DL1 glomerulus that it innervates (arrowhead). (B) Change of fluorescence intensity in the soma and the glomerulus in response to isoamyl acetate. (C) Odor-evoked activity in the soma is highly correlated with that in the associated glomerulus. Each data point represents the integration of $\Delta F/F$ over time. (D) $\Delta F/F$ is plotted against time for the glomerular (Glom) and cell-body (CB) response to six different odorants for two PNs. All odors were applied at 33% SV for 2 s, indicated by the black horizontal lines.

taneously. In each of 18 PNs, a response within the soma was always correlated with activation of the glomerulus that it innervates (Fig. 4D). Cell 1, for example, innervates the DA4 glomerulus, and it responds to hexane with a fluorescence change ($\Delta F/F$) that reaches a peak value of 100% in the dendrites and 24% in the cell body (Fig. 4D). The other five odorants did not elicit detectable fluorescence change in either the dendrites or the soma. A linear regression analysis of activity in glomeruli and cell bodies shows a correlation coefficient of 0.9, indicating that odor-evoked calcium influx in the soma is strongly correlated with glomerular activation. This result suggests that lateral connections do not transform dendritic PN input into a different output. Taken together with our previous result that activity in ORN axons and PN dendrites are highly similar, these data suggest that lateral connections do not significantly alter the receptive field of the PNs.

Relation of Calcium Activity to Action Potentials in PNs. Elevations in intracellular calcium do not necessarily reflect the transmission of action potentials along the PN. We have, therefore, developed procedures to simultaneously perform Ca^{2+} imaging and loose-patch electrical recording of PN cell bodies (Fig. 5A). G-CaMP was expressed in GH146-positive neurons and the preparation was exposed to several odorants. Imaging was used to identify an active PN cell body, and spike activity in this cell body was analyzed by loose-patch recording.

Nine PN cell bodies were identified that were imaging positive to one odorant but not to a second odorant. To determine whether calcium activity corresponds with a robust electrical response, these cells were then subjected to loose-patch recordings. Calcium activity induced by odor stimulation was always accompanied by brisk firing of action potentials with spike frequencies (spikes in the first second of odor response) ranging from 16 to 48 Hz (Fig. 5B Upper). Conversely, the absence of calcium activity was always associated with a few or no action potentials (Fig. 5B Lower). It is worth noting that there were some cases where a small odor response was detected by electrical recording but not by calcium imaging of the cell body, suggesting that the imaging technique is less sensitive than electrical recording. Nonetheless, PN spike output is related to

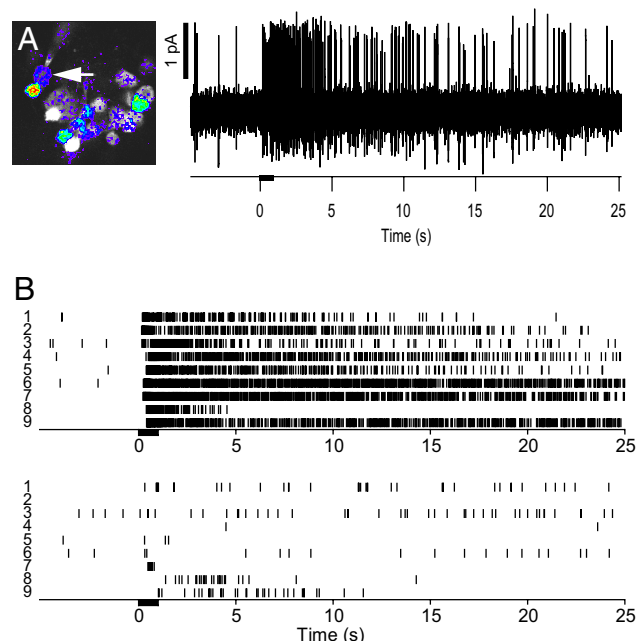


Fig. 5. Calcium imaging correlates with spike firing in PNs. (A) G-CaMP-labeled PNs responsive to odor stimulation were identified by two-photon imaging (Left), and loose-patch recordings were then performed on the identified PN soma (arrow) (Right). (B) Action potentials in imaging-positive PNs (Upper) and imaging-negative PNs (Lower). Cells 1–5 (as shown on the left) were imaging positive for isoamyl acetate and negative for caproic acid. Cells 6–9 were imaging positive for 1-octen-3-ol and negative for isoamyl acetate. Odorants, diluted in saline at 40, 0.2, and 3.6 nM/ml for isoamyl acetate, caproic acid, and 1-octen-3-ol, respectively, were delivered by pressure injection.

somatic calcium activity, which in turn is related to dendritic calcium activity. Thus, our measure of calcium activity in PN dendrites reflects the PN spike output. By examining activity from ORN axons to PN cell bodies, we saw a concordance between each successive level of olfactory processing and did not find evidence that lateral interactions significantly contribute to PN output, indicating that ORN input is the main driving force for PN output.

Population Sparseness of PNs. Our earlier data suggest that PNs are no more broadly tuned than their cognate ORNs. Thus, given the sparse representation at the ORN level (16, 28), we would expect a given odor to excite relatively few PNs. To test this hypothesis, we performed loose-patch recording on randomly chosen PNs and analyzed the population sparseness. We expressed GFP in GH146 PNs and examined the electrical response to two different odorants by loose-patch recordings of 48 randomly chosen cells (Fig. 6A). To quantify odor response we counted the number of spikes in the first second of the response (Fig. 6B). A large population of cells had little or no response (<5 spikes) to either odor. The average number of spikes in response to 1-octen-3-ol was 9, whereas that for isoamyl acetate was 19. For both odors, only two or three cells had a firing rate >2 SD above the average. If firing rate is important for olfactory coding, these briskly firing outliers may be the major carrier of olfactory information. We used the Treves–Rolls sparseness measure (29) to quantify the population distribution of PN responses. An index of 1 indicates that all neurons in the population respond equally, whereas an index near 0 indicates that only a single neuron responds; in other words, the response is extremely sparse. After odor stimulation, the peak sparseness index was 0.19 for 1-octen-3-ol and 0.29 for isoamyl acetate. Isoamyl acetate excites more glomeruli than any other odorant in our collection (13), thus the sparseness index we obtained for this odorant probably represents the upper limit for the population sparseness. Given the fact

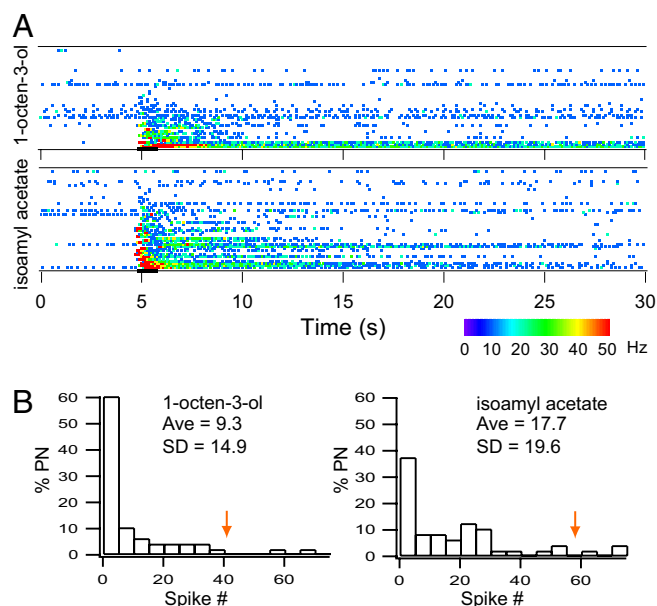


Fig. 6. A sparse population of briskly responsive PNs. (A) Instantaneous firing rate shown in color scale for two different odorants. Bin size = 100 ms. Odorants were diluted in saline at 40 and 3.6 nM for isoamyl acetate and 1-octen-3-ol, respectively, and delivered by pressure injection. Odor concentration was equivalent to 7% SV, determined by a bioassay (see *SI Methods*). (B) Histogram of spike frequency in the first second of odor response for 1-octen-3-ol (Left) and isoamyl acetate (Right). The arrow indicates 2 SD above average.

that the ORN population response is sparse (28), if lateral excitation is a powerful driving force for PN activity, we should see a more distributed odor response in PNs. The sparse population response in PNs is consistent with our findings that ORN receptive fields, rather than lateral connections, are the main driving force for PN output.

Discussion

In this study, we examined propagation of olfactory information in the antennal lobe. We used two different receptor gene mutations to remove all receptor input to specific glomeruli. Silencing the ORNs of the VM2 glomerulus with the *Or43b* mutation allows us to investigate the relative contribution of the cognate receptor input to the PN output activity. Exciting many glomeruli and recording from PNs with no direct receptor input revealed evidence of lateral excitation. However, the cognate receptor input makes a much greater contribution to the PN firing than any interglomerular connections do.

We have extended our investigation of information propagation to many more glomeruli of the fly antennal lobe. First, quantitative imaging experiments showed that odor-evoked calcium activity in PN dendrites was similar to that of ORN axons at the same glomerulus, consistent with the model of glomerular propagation of neural activity from ORNs to PNs. Imaging of single PNs demonstrates correlated calcium elevation in the soma and the associated dendrites, suggesting that lateral connections do not significantly alter PN output. Recording from 48 PNs showed a sparse representation of odors by the antennal lobe.

Our comparison of calcium responses in ORN axons and PN dendrites shows that the receptive field of ORNs is preserved in PNs, and our mutant results show that there is no mechanism for broadening PN tuning. This would seem to contradict a study suggesting that PN tuning curves are much broader than those of their cognate ORNs (16). That conclusion was based on the finding that in some cases, a large PN response can occur when the cognate

ORN response is quite small. However, each glomerulus in *Drosophila*, on average, receives the axons of 30 ORNs and the dendrites of 3 PNs (17). Thus, the PN output may reflect the pooling of all ORN inputs to that glomerulus, which may make each PN more sensitive to odor stimulation than any single ORN. Imaging experiments measure the activity in all ORNs of a single glomerulus; this ensemble activity may more accurately reflect the input to the cognate PNs.

We analyzed ensemble PN activity and found that odors are sparsely represented by the PN population. Our analysis shows that the responses of a few neurons are much greater than the average response. If these robustly responding neurons play an important role in representing olfactory information, the response is indeed very sparse. In contrast, Wilson and colleagues found a very broad PN response (16). In their analysis, spike activity >2 SD above baseline is considered a positive response, which results in the conclusion that a given PN responds to 60% of all odors tested, and a given odor excites 69% of all PNs. Although it remains to be determined what type of PN activity is detected by the third-order neurons, the results from a recent paper by Suh *et al.* (30) suggest that a high firing rate is required to mediate an olfactory behavior in *Drosophila*. These researchers found that there is a tight correlation between the level of ORN activity and the robustness of the CO₂ avoidance behavior. Further experiments addressing what type of PN activity is behaviorally relevant will be crucial to understand odor representation in the antennal lobe.

The results from a recent study by Shang *et al.* (18) have suggested that lateral excitation play a major role in PN activity. In a mutant fly with dramatically lower ORN activity, PNs showed a modest reduction in the odor response. The study by Shang *et al.* used synaptotagmin to measure synaptic release from PN dendrites in the antennal lobe, which may not reflect PN output to higher brain centers. The relationship between PN local synaptic release in the antennal lobe and PN spike activity remains to be determined. While this work was in review, another paper investigating the role of lateral connections in the antennal lobe was published (31). The findings of Olsen *et al.* (31) are qualitatively similar to ours, in that PN output is dramatically reduced in the absence of direct ORN input. However, their study consistently showed a more robust PN response in both wild-type and mutant flies. This finding can be attributed to several differences in recording conditions. First, we used the loose-patch recording technique, whereas they used the whole-cell-patch method. Until the neuronal electrolyte composition is known, the more invasive whole-cell recording method may alter the excitability of the cell. Second, differences in PN responsiveness may be because of different preparations; Olsen *et al.* used a whole-fly preparation, whereas we used an antennae-brain preparation that maintains only the olfactory sensory input. The antennae-brain preparation allows for greater accessibility for optical and electrical recording, but the lack of nonolfactory sensory input and hormonal modulation may affect antennal lobe excitability. On the other hand, the stress associated with the dissection and immobilization required for the whole-fly preparation may alter antennal lobe excitability as well. Third, for the electrical recordings, we applied odorants in the liquid phase instead of gas phase, which may affect ORN responsiveness. Finally, differences in saline composition may also affect excitability; for example, Olsen *et al.* used 130 mM sodium, whereas we used 113 mM. Despite the fact that our experimental conditions lead to quantitative differences in the numbers of spikes, these two studies both support the notion that cognate ORNs are the main source of PN activity, whereas lateral excitatory connections make a relatively small contribution to the response. Further experiments will be required to determine whether lateral activity is read by the third-order neurons and contributes to behavioral output.

Implications for Olfactory Processing. Receptor activation by odorants results in spatial patterns of activity in the *Drosophila* antennal

lobe and constitutes the main driving force for PN excitation. These patterns are modulated by interglomerular interactions, resulting in PN action potentials that are ultimately conveyed to higher olfactory centers. This conceptual framework is reminiscent of the notion of drivers and modulators in the mammalian visual system (32, 33). Drivers are defined functionally as the transmitters of the receptive field, whereas modulators can alter the efficacy of the transmission, without altering its basic properties. Drivers typically project from one layer of a sensory system to another, whereas modulators are confined to a single layer. The anatomy of the antennal lobe and our results suggest that ORNs can be seen as the drivers of PNs and interneurons as the modulators.

The finding that PNs receive excitatory as well as inhibitory input from interneurons raises the question of what role these lateral interactions play in olfactory coding. Lateral inhibition has been proposed as a mechanism for olfactory processing, either by sharpening odor tuning (34, 35) or by generating synchrony (36, 37). One plausible function of lateral excitation, in concert with lateral inhibition, is to enhance PN synchrony. Synchronized activity could facilitate the readout of the combinatorial code by the third-order neurons (38). Another possible function could be to regulate PN sensitivity. Concurrent excitation and inhibition could be useful for fine-tuning neuronal responsiveness. Indeed, it has been demonstrated that injecting more balanced excitatory and inhibitory inputs can act to reduce the sensitivity of pyramidal neurons, whereas a reduction in the amount of the balanced input can increase sensitivity (39). It is tempting to speculate that balanced inputs in the antennal lobe could have a similar role in modulating PN responsiveness, for example to allow these neurons to respond optimally to a wide range of odor concentrations. These balanced inputs could also mediate top-down modulation of PN sensitivity to reflect the behavioral state of the organism.

Materials and Methods

Experimental Animals and Preparations. Flies were raised on standard medium at 22–25°C. See *SI Methods* for genotypes and sources of flies used.

The isolated brain preparation for imaging was described in ref. 13. For electrophysiology, the dissecting procedure was the same as for the imaging preparation, except that the sheath covering the antennal lobes was mechanically removed by using a pair of fine forceps. After dissection, the preparation was rinsed and kept in adult hemolymph-like (AHL) saline.

Odor Stimulation. The olfactometer used to deliver airborne odors was described in ref. 13. Odor concentrations are specified for each experiment. In all electrical recording experiments, solutions of odorants in saline were applied to the antenna submerged in saline with a dual-channel pressure injection system (Pressure System II; Toohey Company, Fairfield, NJ) by a micropipette of 2- μ m tip diameter at 5 psi (1 psi = 6.89 kPa) for 1 s.

Calculation of Population Sparseness. The Treves–Rolls sparseness is defined as:

$$S = [\sum r/N]^2 / \sum [r^2/N],$$

where r is the firing rate of a given cell at a given 100-ms time bin, and N is the number of cells.

Electrical Recordings and Calcium Imaging. Calcium imaging was performed with a custom-built two-photon microscope as described in ref. 13. Loose-patch and bristle recordings were performed with an EPC-7 amplifier (Heka Elektronik, Southboro, MA). The resistance of the saline-filled electrodes was typically 6–10 M Ω . See *SI Methods*.

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